QUESTION 1

(a)

Using relationship optionality; mandatory one, mandatory one to many, optional zero to one and optional zero to many, draw the EERD of the following;

MadraGra (identified by id and location) is an organisation which has two centres; MG-A and MG-B.

MG-A and MG-B each have a distinct eircode.

MadraGra plans on establishing a third centre at some time in the future.

MadraGra employs three types of employees only; Subject Matter Experts (SMEs), Tutors and Developers. Sometimes an SME is also a tutor or developer. All employees have an employee id, name, DOB (date of birth), start-date and contact details. The three types of employee each hold additional information; an SME has a topic speciality, the Developer has number of years experience and the Tutor has number of courses.

All employees have a particular Job Status; Part time or Full time. There will be no new job status types. Also note that an employee who is part time cannot also be full time.

The only information held with Job Status is the Status ID.

Part time includes pay per hour, whilst Full time includes Salary.

All Tutors are part time. All SMEs and Developers are full time.

Students register with MadraGra. The student is described by their ID, name, DOB, nationality and contact. A student attends 1 or more courses. A course has to have at least one student. A course is identified by Course ID, description and recommended reading.

A course can be taught by 1 or more tutors but has to have at least one student registered to run.

A tutor can teach one or more courses. Each course is developed by at least one developer.

Developers can develop many courses.

(16 marks)

(b) Using examples, discuss at least three reasons why databases are more powerful than file based systems. **(9 marks)**

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QUESTION 2

(a) – all parts are worth equal marks

EMPLOYEE

StaffNo	FName	LName	Position	Gender	DOB	Salary	BranchNo
SL21	John	White	Manager	M	1-Oct-1945	89000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-1988	45000	B003
SG14	David	Ford	Supervisor	M	24-Mar-1972	60000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-1978	39000	B007
SG5	Susan	Brand	Manager	F	3-Jun-1980	82000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-1990	31000	B005

Given the table **EMPLOYEE**, write the SQL which;

1. Counts the number of employees in each branch and displays the following;

Number of employees per branch
There are 3 employees in B003 There are 2 employees in B005 There are 1 employees in B007

2. Produces a list of the top 3 salaries of all staff, showing staff number, first and last name, DOB and salary, where these salaries are between the minimum and average of ALL salaries (hint:subquery).

3. Displays the sum and average of all salaries based on position [note: the data is displayed in descending order of the sum of the salaries].

Sum of Salaries	Average of Salaries	Position
171000	85500	Manager
115000	3833.33	Assistant
60000	60000	Supervisor

4. Displays the following salary information based on position;

Salary spend
The total salary spend on managers was 171000. The max was 89000 and the average was 85500. The total salary spend on assistants was 115000. The max was 45000 and the average was 38333.33. The total salary spend on supervisors was 60000. The max was 60000 and the average was 60000.

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5. Counts the number of first names which contain the letter a. Categorise by gender and display the following;

Results
There are 3 first names with an a – grouped by gender = F
There are 1 first names with an a – grouped by gender = M

6. Displays the following for all records where the salary * 1.2 is greater than ALL of the list of results returned when looking for the average salary where the last name contains an E, grouped by position;

StaffNo	FName	LName	Position	Gender	DOB	Salary	BranchNo
SG5	Susan	Brand	Manager	F	03-Jun-80	82000	B003
SL21	John	White	Manager	M	01-Oct-45	89000	B005

(15 marks)

(b) Given the relations CUSTOMER and CUSTOMER ORDER, write the sequence of operators in **relational algebra** to extract data to satisfy the following information needs (a-d) – **all parts are worth equal marks**.

CUSTOMER

CustID	Name	Phone	Status
C1	Anne Boleyn	086 7238904	Frequent
C2	Sean Fitzpatrick	087 5534666	Occasional
C3	Mary Hunes	088 7543588	Occasional
C4	Tom Sawkes	086 6234765	Occasional
C5	Liam Luoin	087 7567378	Frequent
C6	George Sunders	088 3464728	Occasional
C7	Tina Hunt	088 4575838	Frequent

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CUSTOMER ORDER

CustID	OrderId	Product	Price	Qty
C1	O1	Doll-Small	12.99	2
C1	O2	Playhouse	14.55	1
C2	O1	Tricycle	56.99	1
C6	O1	Doll-Big	24.87	2
C6	O2	Laptop	290.87	1
C6	O3	Bicycle	356.76	1

a. Show only the product and its associated price where the price is between 30 and 190.

b. Display the customer ids (CustID) of those who didn't place an order. Rename CustID to 'Check Customers' and the resulting relation to 'NO ORDERS'.

c. Display the average, maximum and minimum price of a customer order grouped by quantity (qty).

d. Write the resulting relation based on CUSTOMER =⋈ CUSTOMER ORDER.

(10 marks)

QUESTION 3

(a) Explain why the following STUDENT relation is in 1NF and not in 2NF. Bring it to 2NF following the 5 step process.

STUDENT

StudentID	StudentName	ProfID	ProfName	Grade
1	Tom	1	Smith	A
1	Tom	2	Jones	B
2	Mike	1	Smith	C
3	Alan	1	Smith	A

(13 marks)

(b) Using examples, discuss what is meant by the term ACID.

(8 marks)

(c) Using examples, explain what is meant by the terms Data Dependence and Data Abstraction.

(4 marks)

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Compare and contrast file, record, field and primary key, their importance

QUESTION 4

(a) Explain why the SEMESTER table is not in 3NF. Bring it to 3NF following the 3 step process.

SEMESTER

Course	Semester	#Places	TeacherID	TeacherName	Room	Location
IT101	2021-1	100	332	Alan Jones	R231	East Wing
IT101	2021-2	120	332	Alan Jones	R231	East Wing
IT102	2021-1	220	495	Tim Bentley	R233	West Wing
IT102	2021-2	150	332	Alan Jones	R234	South Wing
IT103	2021-2	140	242	Mary Smith	R234	South Wing

(12 marks)

(b) Using examples, explain the following; Entity Integrity, User Defined Integrity, Domain Integrity and Referential Integrity.

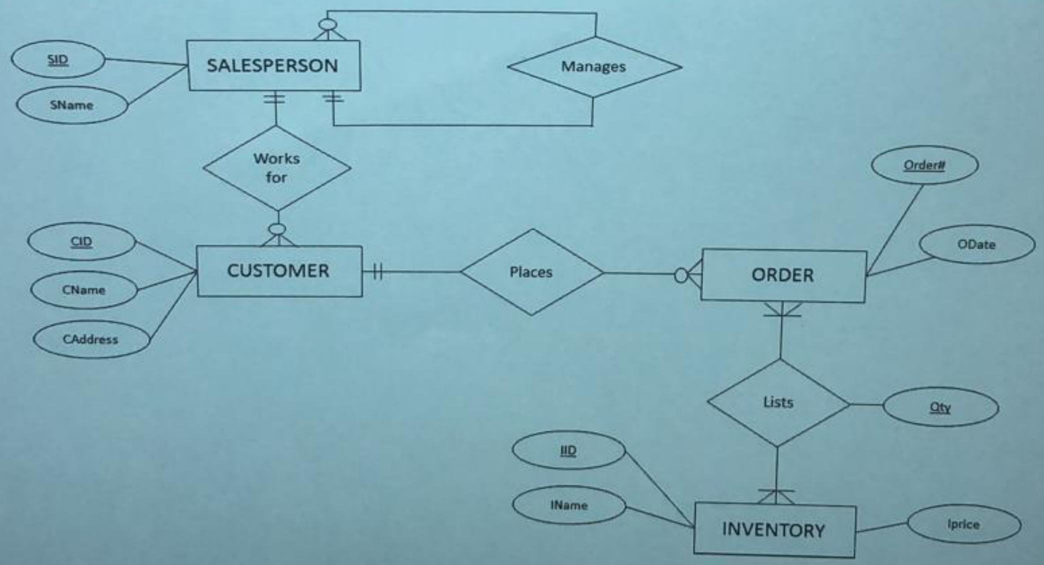
(8 marks)

(c) *Using examples, explain the following;* Data Atomicity and Operational Databases.

(5 marks)

QUESTION 5

(a) Using the **graphical approach,** map the following to relations. ***(10 marks)***



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(b) Using examples, discuss the stages of the System Development Life Cycle for Database Development. **(8 marks)**

(c) Perform a vertical partition on the following COURSE table given that TeacherID, Room and Location are rarely queried.

COURSE

Module	#Places	TeacherID	TeacherName	Room	Location
IT101	100	332	Alan Jones	R231	East Wing
IT102	220	495	Tim Bentley	R233	West Wing
IT103	150	332	Alan Jones	R234	South Wing
IT104	140	242	Mary Smith	R234	South Wing

(7 marks)

QUESTION 6

(a) Explain the purpose and different layers of the 3-Schema Architecture. **(10 marks)**

(b) Given the following tables; PRODUCT and ORDERPRODUCT;

PRODUCT

PID	Description	Price
P1	Toaster	12.99
P2	Book	9.99
P3	Material	19.99
P4	Bicycle	120
P5	Drone	245.67

ORDERPRODUCT

OrderID	PID	Qty
O1	P1	4
O1	P2	3
O2	P1	5
O2	P5	3
O3	P2	3
O3	P4	2
O4	P1	6
O4	P2	5

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Write the SQL which;

1) Displays the sum of the price times (x) quantity categorised by order as follows;

OrderID	Sum : Price x Quantity
O1	81.93
O2	801.96
O3	269.97

2) Revise (1) and only include those order IDs where the sum of price times (x) quantity is greater than 200.

(8 marks)

c)
In the COMPONENT table, a dependency was found to exist between a non-key attribute (REF#) and a key attribute (ItemID). Bring the COMPONENT table to BCNF, by firstly drawing the functional dependency diagram to reflect this new dependency, mapping to relations using the graphical approach and then redrawing the tables.

COMPONENT

ItemID	Part#	Description	Cost	REF#
1054	A	Cotton Bag	39.99	015312
1054	B	Metallic zipper	2.99	015312
1055	A	Cotton Purse	5.99	015372
1056	A	Cotton Hat	12.99	015373

(7 marks)